Business Understanding Report

Loan approval prediction: preparing application data for machine learning

Prepared by	Cheruiyot Nickson Kirui	Programme	AnalystLab Africa Data Science Internship, Week 2
Dataset	Loan Prediction Problem (Kaggle)	Date	October 2026

1. Business problem

Dream Housing Finance is a home-loan company with a presence in urban, semi-urban and rural areas. Customers first apply for a loan, and the company then checks whether they are eligible, using details such as gender, marital status, education, number of dependents, income, loan amount and credit history. Checking each application by hand is slow, and different reviewers can reach different decisions on similar applications. The company wants to automate the eligibility check using the information applicants provide when they apply.

2. Project objective

The objective of this project is to **prepare the application data so that a classification model can be trained on it**: cleaned, consistently encoded, scaled and free of data leakage, with every preprocessing decision documented. The model itself is the next stage. This week delivers a machine-learning-ready dataset and the reasoning behind it.

3. Target variable

The target is **Loan_Status**: Y (approved) or N (rejected). The labelled file holds **614 applications**, of which 68.7% were approved and 31.3% rejected. For modelling it is encoded as **loan_approved** (1 = approved, 0 = rejected), which makes this a **binary classification** problem with moderately imbalanced classes. A second, unlabelled file of 367 applications is used only to prove the preparation pipeline works on new data.

4. Importance of predictive modelling

- **Speed:** a model can score an application the moment it is submitted, instead of waiting for manual review.
- **Consistency:** the same criteria are applied to every applicant, reducing differences between reviewers.
- **Risk control:** approving applicants who cannot repay creates losses, while rejecting good applicants loses business. A model helps balance both.
- **Staff focus:** clear-cut cases can be handled automatically, leaving people to review borderline applications.
- **Insight:** modelling shows which factors drive decisions, which helps the company check that its policies are applied as intended.

5. Expected business impact

A reliable model is expected to shorten approval times, make decisions more consistent, reduce risky approvals and let the company handle more applications without proportionally more staff. These benefits should be measured against clear indicators once a model is in use: average time to decision, the share of applications decided automatically, the repayment performance of approved loans, and how well the model catches applications that should be rejected.

6. What the data already tells us

- **Credit history is the dominant signal.** Applicants whose credit history meets guidelines are approved about 79% of the time, against about 9% for those whose history does not.
- **Data is incomplete:** 22% of applications have at least one missing value, and 8% have no credit history recorded.
- **Incomes and loan amounts are heavily skewed**, with a few very large values that need careful handling.

- **Accuracy alone is misleading:** approving everyone would already be right 69% of the time, so models must be judged on how well they identify rejections.

7. Risks and responsible use

- **The target is a past decision, not a repayment outcome.** A model trained on it learns how the company decided before, including any bias in those decisions, and not whether borrowers repaid.
- **Fairness:** gender and marital status can lead to discriminatory outcomes and are restricted in credit decisions in many places. They are excluded from the final dataset. Property area could act as a stand-in for location and should be reviewed before real use.
- **Small sample:** 614 labelled rows means results can vary a lot, so models should be tested with cross-validation.
- **Human oversight:** the model should support loan officers, especially for borderline applications, and not replace them.